Daily Report — July 22, 2026

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Yesterday's Session (July 21) — Where We Stopped

Worked on the **ECG Signal Preprocessing Notebook** (Offline Task 1) with the MIT-BIH Arrhythmia Database. Final notebook state:

- ✓ Data loading & inspection: all 48 MIT-BIH records loaded via `wfdb`, MLII channel selected as primary.
- ✓ FFT frequency analysis — confirmed key ECG bands (P wave: 0.67–5 Hz, QRS: 10–50 Hz, T wave: 1–7 Hz).
- ✓ Seven filtering methods implemented and compared: HP Butterworth, Notch, Moving Average, Savitzky–Golay, Chebyshev II, SWT, EMD.
- ✓ **Winner selected:** SWT with `rbio3.9` wavelet, level 5, threshold scale 0.5 (correlation: 0.9989, PRD: 2.26 %) — consistent with Ádám et al. (2025).
- ✓ **Optimal pipeline locked:** HP Butterworth (0.5 Hz, order 4) → Notch (60 Hz) → SWT.
- ✓ Beat segmentation using AAMI label mapping → 105,039 beats across N/S/V/F/Q classes.
- ✓ Per-beat z-score normalisation applied.
- ✓ Patient-level train/test split: 83,736 train (36 patients) / 21,303 test (9 patients), zero patient overlap.
- ✓ Output saved: `mitbih_preprocessed.npz`.

Key Decisions

- Three-way split abandoned → 80/20 patient-level split (small patient pool caused entire classes to disappear from validation).
- Class imbalance (86 % Normal) deferred to model stage via class-weighted loss functions.
- Records 201 and 202 always grouped together (same patient).
- Downstream model/task not yet determined.

Today's Plan — July 22

Priority 1: Literature Search (Catch-Up)

- ○ Resume paper search for the mini review — currently at 5/8 papers.
- ○ Focus on filling gaps: need papers on hybrid DL+chaos approaches and chaos-based ECG features.
- ○ Use Connected Papers starting from the NG-RC (Gauthier) and ECG (Ádám) papers as seeds.
- ○ For each new paper found: immediately fill all 11 Excel columns.
- ○ Verify Q-rankings and impact factors on SJR for all papers.

Priority 2: Novelty / Gap Check

- ○ Search: "Has anyone applied NG-RC or RC to ECG classification?"
- ○ Search: "Inter-patient vs. intra-patient splitting for ECG classification — impact on model generalisation?"

- Search: "Lyapunov exponents / correlation dimension as features for DL-based ECG arrhythmia detection?"
- Search: "Hybrid RC + LSTM / RC + Transformer architectures for chaotic time series?"
- Document findings — if a direction is already well-explored, note it and pivot.

Priority 3: Notebook Next Steps (If Time)

- Load `mitbih_preprocessed.npz` and build baseline classifiers (1D-CNN, LSTM) with class-weighted loss.
- Explore whether extracting chaos features (Lyapunov, entropy, D_2) from individual beats improves forecasting.
- Lower priority than finishing the literature search today.

Status: ■ Behind on literature search. Preprocessing pipeline done. Model stage not started.